P510/1
PHYSICS
Paper 1
July. /Aug. 2026
3 hours

WAKISO-KAMPALA TEACHERS' ASSOCIATION (WAKATA)

WAKATA MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

PHYSICS

WAKATA 2026

Paper 1

3 hours

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two** *sections;* **A** *and* **B**. *It has* **six** *examination items.*

Section **A** *has* **two** *compulsory items.*

Section **B** *has* **two** *parts;* **I** *and* **II**. *Respond to* **one** *item from each part.*

Respond to **four** *examination items in all.*

Any additional item(s) responded to will **not** *be scored.*

All *responses* **must** *be written in the answer booklet(s) provided*

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

Neat work is a must!!

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SECTION A

This section is **compulsory**

Item 1.

At a mechanical engineering workshop at MPoma Satellite, Engineer Massy and his Senior Six student, Hilly, are studying the elastic behaviour of metals for designing a precision suspension system.

Two wires, one made of steel and the other made of bronze, are to be considered for use. One of them must be selected because it has the lowest extension when a load of 280 N is attached to its free end. The steel wire has a length of 0.55 m and a cross-sectional area of $1.2 \times 10^{-6} m^2$. It is connected to a bronze wire of length 0.42 m and cross-sectional area of $2.2 \times 10^{-6} m^2$.

Engineer Massy selected another wire to pair with the chosen wire, but he could not determine its Young's modulus to ensure that the combination works effectively in the workshop.

Later, Hilly takes a small metal ball of mass 0.45 kg and attaches it to a light, inextensible string of length 0.9 m. In an attempt to hit a bird in the compound, he whirls the ball in a horizontal circle of radius 50 cm at a constant speed of 5.5 m/s before releasing it. The bird is 20 m away from him. If the bird flies for 4.5 s immediately after the ball is released, Hilly wonders whether the ball will hit the bird.

Finally, a technician at a satellite ground station becomes concerned about a communication satellite orbiting the Earth at an altitude of 35,000 km. Hilly wonders, "What is the orbital period of this satellite? Is it in a geostationary orbit?"

Mary, Hilly's friend, could not explain why satellites in lower orbits move faster than those in higher orbits and how this principle is useful in communication.

Support Material

Young's modulus of steel $= 2.0 \times 10^{11} Pa$
Young's modulus of bronze $= 1.0 \times 10^{11} Pa$
Acceleration due to gravity, $g = 9.8 m/s^2$
Earth's radius $= 6.4 \times 10^{6} m$
Gravitational constant, $G = 6.67 \times 10^{-11} Nm^2/kg^2$
Mass of Earth $= 6.0 \times 10^{24} kg$

Task:

(a) Study the workshop situation carefully and guide Hilly in addressing all his concerns.

(b) Address Mary's concern.

Item 2.

At a certain school, the Physics teacher, Mr Luli, and his Senior Six student, Joan, are investigating a cathode ray tube (CRT) in the laboratory.
When they switch on the tube, a bright green glow appears on the fluorescent screen. Joan wonders, "What are cathode rays? Are they particles or waves?"
Mr Luli could not explain why the rays are deflected by a magnetic field but not by an electric field when the tube is incorrectly set up. Joan becomes worried that their apparatus may be faulty.

To understand the phenomenon better, Mr Luli asks Joan to calculate the specific charge $(e/m)$of cathode ray particles using Thomson's method. Electrons accelerated through a potential difference of 750 V enter a magnetic field of 0.032 T and move in a circular path of radius 6.0 cm. Joan wonders whether her answer will agree with the known value of $1.76 \times 10^{11} C/kg$.

Mr Luli then demonstrates the production of X-rays using a Coolidge tube. He explains that fast-moving electrons are suddenly stopped by a metal target, producing X-rays. Joan wonders, "What determines the minimum wavelength of the X-rays produced? Why are X-rays dangerous to living tissues?"
Mr Luli becomes concerned because Joan cannot explain how X-rays are used in medical imaging while also being harmful.

Mr Luli then draws Bohr's atomic model on the whiteboard. He explains how electrons move between energy levels by absorbing or emitting photons.
Joan wonders, "Why does hydrogen produce a line spectrum while hot solids produce a continuous spectrum?" She also cannot explain why Bohr's model fails to fully explain atoms such as helium and carbon.
Finally, Mr Luli asks Joan to calculate the ionisation energy of hydrogen using the Rydberg constant:

$$R = 1.097 \times 10^{7} m^{-1}$$

Joan worries whether her answer in electron volts will agree with the known value of 13.6 eV.
Martha, a friend to Joan, wonders whether electrons emitted through thermionic emission can be accelerated into a magnetic field to determine the specific charge of electrons experimentally.

Support Material

Electron charge, $e = 1.60 \times 10^{-19} C$
Electron mass, $m = 9.11 \times 10^{-31} kg$
Accelerating voltage $= 750\ V$
Magnetic field strength, $B = 0.032T$

Turn Over

Radius, $r = 6.0cm = 0.060m$
Planck's constant, $h = 6.63 \times 10^{-34} Js$
Speed of light, $c = 3.0 \times 10^{8} m/s$

$$1eV = 1.60 \times 10^{-19} J$$

Rydberg constant, $R = 1.097 \times 10^{7} m^{-1}$
Known specific charge of electron $= 1.76 \times 10^{11} C/kg$

Task:

(a) Examine Joan's observations and assist her in resolving all her challenges.

(b) Assess Martha's concern.

SECTION B

PART I

Respond to only **one** *item from this part*

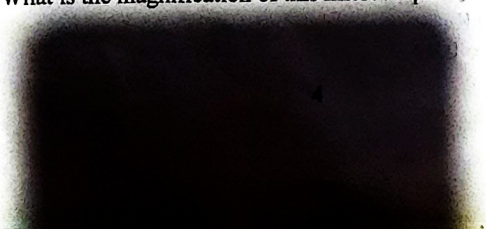
Item 3.

At a rural health centre, a doctor and his intern, Peter, are facing challenges with a damaged solar-powered vaccine refrigerator.
The refrigerator operates using a thermodynamic cycle to remove heat from the vaccine compartment, which is maintained at 5°C, and releases the heat to the surrounding air at 35°C.

The compressor does 600 J of work per cycle, while the refrigerator absorbs 2,000 J of heat from the cold compartment during each cycle.
Peter wonders, "How much heat is released into the room? Is this refrigerator operating as efficiently as it should?"

He calculates the coefficient of performance (COP) but becomes concerned because the actual performance is lower than the ideal Carnot performance.
A broken window allows humid air from outside, with a relative humidity of 80% at 35°C, to enter the refrigerator. The doctor explains that water vapour in the air condenses on the cold evaporator coils, causing ice formation.

Peter cannot explain why ice forms even though the surrounding air temperature is above the freezing point of water. The doctor reminds him about the concepts of latent heat of vaporisation and condensation.
To inspect the vaccine vials, Peter uses a compound microscope with an objective lens of focal length 0.6 cm, an eyepiece lens of focal length 2.5 cm, and a tube length of 11 cm.
He wonders, "What is the magnification of this microscope? Why is the image inverted?"



Dr Sembatya then asks Peter to use a small refracting telescope with an objective focal length of 70 cm and an eyepiece focal length of 7 cm to observe a distant ambulance across the valley.

Peter becomes uncertain because he cannot remember whether a telescope in normal adjustment produces an upright or inverted image.
Later, Peter helps repair the solar panel that powers the refrigerator. The panel produces 280 W of power for 4.5 hours of sunlight each day.

He calculates the energy produced and wonders, "Is this enough to operate the refrigerator if the compressor runs for 30% of the time?"
The compressor consumes 200 W when running.
After Peter narrates his experience to a friend, the friend wonders whether there is a system that can be used to determine the specific latent heat of vaporisation experimentally.

Support Material

Saturation vapour pressure at 35°C = 5.62 kPa
Saturation vapour pressure at 0°C = 0.61 kPa
Latent heat of vaporisation = $2.26 \times 10^6 J/kg$

Task:

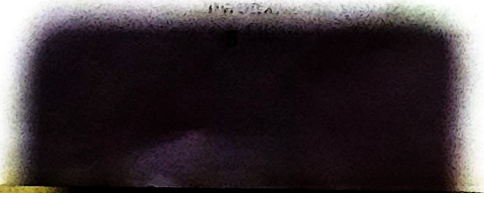
(a) Analyse the health centre situation and assist Peter in addressing all his concerns.

(b) Answer Peter's friend's concern.

Item 4.

At a secondary school in Mbale, Physics teacher Mr Mutonyi and his Senior Six student, Fiona, are investigating heat conduction through different materials.
They have a metal rod of length 65 cm and cross-sectional area $1.2cm^2$. One end of the rod is placed in boiling water at 100°C while the other end is placed in melting ice at 0°C.
The thermal conductivity of the metal is $170W/(mK)$.
Fiona wonders, "What is the rate of heat transfer through the rod? How much ice melts per minute?"

She cannot explain why a copper rod of the same dimensions would transfer heat faster.
Mr Mutonyi then asks Fiona to consider a heat engine operating between a hot reservoir at 500°C and a cold reservoir at 30°C.
The engine absorbs 3,800 J of heat per cycle and performs 1,400 J of work.
Fiona wonders, "What is the actual efficiency of this engine? How does it compare with the Carnot efficiency?"



She becomes concerned because the actual efficiency is much lower than the ideal efficiency.
Later, Fiona investigates the expansion of an ideal gas.
A gas at 260 K occupies $0.035 m^3$ at a pressure of $1.2 \times 10^5 Pa$. It is heated at constant pressure until its volume doubles.

Fiona wonders, "What is the final temperature? How much work is done by the gas?"
She cannot explain why the internal energy of the gas increases during the process.
Finally, Bena, another student, investigates stationary waves on a string. A string of length 1.2 m is fixed at both ends and vibrates in its fourth harmonic at a frequency of 200 Hz.
Fiona wonders, "What is the speed of waves on the string? What is the fundamental frequency?"
She cannot explain why the string vibrates with nodes and antinodes.

Support Material
Specific latent heat of fusion = $3.34 \times 10^5 J/kg$

Task:

(a) Examine the laboratory investigation and help Fiona resolve all her challenges.

(b) Analyse Bena's investigation and describe the relationship between frequency and tension in a vibrating string.

SECTION B

PART II

Respond to only **one** *item from this part*

Item 5.

At a printing press in Kampala, an inkjet printer is used to produce high-quality documents. The printer forms images by ejecting tiny charged ink droplets. These droplets pass between two parallel plates at a potential difference of 550 V, where an electric field deflects them to the correct positions on the paper.
For the ink droplets to be properly directed, the force acting on each droplet must be greater than $2.8 \times 10^{-6} N$.

The charging system is powered by a DC supply of 160 V through a resistor R that controls the current. According to the printer specifications, a current of 0.22 A should flow through the resistor for the charging system to operate correctly.
However, during operation, the printer becomes hot and produces faint, poorly aligned prints. A technician is called to investigate the problem.
After opening the printer, the technician discovers that the resistor is damaged and requires replacement. The resistor was made from a wire of a certain resistivity, but the exact length of wire required and its resistivity are unknown.
The technician wonders, "How long should the wire be?"

He is also concerned about the electric deflection system of the printer.

Support Material

Area of deflecting plates = $4.0 \times 10^{-3} m^2$
Separation between plates = $0.8 \times 10^{-4} m$
Charge on one droplet = $3.5 \times 10^{-13} C$
Permittivity of free space, $\epsilon_0 = 8.85 \times 10^{-12} F/m$
Resistivity of wire = $1.0 \times 10^{-6} \Omega m$
Diameter of wire = *0.08 mm*
DC supply voltage = *160 V*
Required current = *0.22 A*

Task:

(a) Analyse the printer situation and help the technician resolve all his challenges.

(b) Explain how the resistivity of the wire can be determined.

Item 6.

In a community, people listen to radios for various reasons, including emotional comfort, mental stimulation and keeping informed about current events.
John visits his parents together with his child, who is studying Physics in Senior Six. During the visit, John's parents complain that their radio set often fails to tune to their favourite religious programme, which operates at a frequency of 0.755 MHz.

John opens the radio and discovers that the tuning circuit, which consists of a capacitor and an inductor, is weak. He decides to replace the components with stronger ones.
He purchases materials and constructs two small capacitors. Each capacitor has a plate separation of 2 mm, a maximum overlapping area of $0.005m^2$, and is filled with a material having a dielectric constant of 4.

The two capacitors are connected in series and combined with an inductor to form the tuning circuit.
John also makes a pure inductor using an air-cored solenoid of length 15 cm, with 100 turns and a cross-sectional area of $0.012m^2$.
John asks his child to perform the necessary calculations to determine whether the new circuit can tune to the desired radio channel.

John's parents also use a radio charging device and ask him to explain its structure and mode of operation. They have always wondered how a charger connected to a socket labelled 240 V can supply a radio that requires only 12 V.

Support Material

Permittivity of free space: $\epsilon_0 = 8.85 \times 10^{-12} F/m$
Dielectric strength of air: $3.0 \times 10^{6} V/m$

Task:

As a Senior Six Physics student:

(a) Analyse the scenario and help John address the challenges with the radio tuning system.

(b) Assess the concern of John's parents about the structure and operation of the charging device.

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END